

Artin's conjecture for ρ_{1951} under GRH and under Selberg's orthonormality conjecture

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Abstract

ρ is the even icosahedral Artin representation of conductor 1951, and Artin's conjecture for it remains open. This note asks what standard conjectures would give. Under the Generalized Riemann Hypothesis every pole of $L(\rho, s)$ lies on the critical line, at a zero of an explicit degree-10 Hecke L -function. With Booker's theorem, either Artin's conjecture holds for ρ , or $L(\rho, s)$ has infinitely many poles there. We show that GRH cannot do more by itself: the obstruction found in the companion paper lives on the critical line and is consistent with GRH. By contrast, Selberg's orthonormality conjecture (Conjecture B) implies Artin's conjecture for ρ ; this implication is, as far as we know, due to M. R. Murty, and we give the argument in full. We then sharpen it: orthonormality among the primitive divisors of two explicit Hecke L -functions, of degrees 12 and 10, already implies Artin's conjecture for ρ , with the sign step done by a degree count. The comparison shows what information a proof needs: averages of Frobenius traces over primes, not locations of zeros.

1 Setting

$G = \text{Gal}(N/\mathbb{Q}) \cong \text{SL}(2, 5) \times C_5$ and $\rho = 2 \boxtimes \psi^3 = \rho_0 \otimes \chi^3$, where χ is a primitive quintic character mod 1951. The irreducibles of $\text{SL}(2, 5)$ are written $1, 2, 2', 3, 3', 4, 4', 5, 6$, with $4 = \text{Sym}^3 \rho_0$ and $4' = \rho_0 \otimes \rho'_0$. We use three unconditional facts from the companion paper.

- (F1) *Brauer formula.* $\rho = \text{Ind}_{\text{Dic}_5 \times C_5}(\tau_1 \boxtimes \psi^3) - \text{Ind}_{\text{Dic}_3 \times C_5}(\mu \boxtimes \psi^3)$, that is, $L(\rho, s) = L_{12}(s)/L_{10}(s)$. Here L_{12} and L_{10} are Hecke L -functions of finite-order characters over number fields of degrees 12 and 10, with Galois sides $(2 + 4 + 6) \otimes \chi^3$ and $(4 + 6) \otimes \chi^3$. Both are entire.
- (F2) *Barrier.* Let $\theta^* = 6 \otimes \chi^3 - \rho$. Then $\langle \theta^*, X \rangle \geq 0$ for every character X induced from a subgroup not containing $\text{SL}(2, 5) \times 1$, and for every tensor product of irreducibles outside the twist class of ρ . On the other hand $\langle \theta^*, \rho \rangle = -1$.
- (F3) *Monomiality.* $6 \otimes \chi^3 = \text{Ind}_{\text{Dic}_5 \times C_5}(\mu_1 \boxtimes \psi^3)$, so $L(6 \otimes \chi^3, s)$ is an entire Hecke L -function.

We also use Booker's theorem [1]: a 2-dimensional irreducible Artin representation of $G_{\mathbb{Q}}$ whose L -function has only finitely many poles is automorphic. In particular its L -function is then entire.

2 What GRH gives

By GRH we mean that every Hecke L -function of a finite-order Hecke character of a number field has all its nontrivial zeros on $\text{Re } s = \frac{1}{2}$.

Theorem 1 ([T|GRH]). *Assume GRH. Then:*

1. *every pole of $L(\rho, s)$ lies on $\operatorname{Re} s = \frac{1}{2}$, at a zero of L_{10} , and its order is at most the order of that zero;*
2. *either Artin's conjecture holds for ρ , or $L(\rho, s)$ has infinitely many poles on $\operatorname{Re} s = \frac{1}{2}$;*
3. *the number of poles with $0 < \operatorname{Im} s \leq T$ is at most $N_{L_{10}}(T) = \frac{10T}{2\pi} \log T + O(T)$.*

Proof. 1. By (F1), $L(\rho, s) = L_{12}(s)/L_{10}(s)$ with L_{12} entire, so a pole of order m at s_0 forces L_{10} to vanish to order at least m at s_0 .

- For $\operatorname{Re} s > 1$, L_{10} is a convergent Euler product, so it has no zeros there.
- On $\operatorname{Re} s = 1$ it is nonzero (Hecke).
- For $\operatorname{Re} s < 0$, the zeros of the completed function are excluded by the functional equation. The trivial zeros of L_{10} are cancelled by the poles of its Gamma factors, which are the same as those of L_{12} and $L(\rho)$ on the real axis to the left. Precisely: $\Lambda(\rho) = \Lambda_{12}/\Lambda_{10}$, and the completed functions are entire with all zeros in $0 < \operatorname{Re} s < 1$.

So poles of $\Lambda(\rho)$ lie at nontrivial zeros of L_{10} , which lie on the line by GRH. The Gamma factor of ρ has no zeros, so the poles of $L(\rho)$ in $0 < \operatorname{Re} s < 1$ are those of $\Lambda(\rho)$.

2. If the set of poles is finite, Booker's theorem gives automorphy, hence entireness.
3. This follows from item 1 and the Riemann–von Mangoldt formula for L_{10} (degree 10, conductor 1951^8).

□

3 Why GRH cannot do more by itself

We make precise the statement that no argument that uses only orders of zeros and poles, and their locations, can derive Artin's conjecture for ρ from GRH.

Definition 2. An *order assignment* at s_0 is an integer vector $n = (n_\pi)$ indexed by the irreducible characters of G , read as $n_\pi = \operatorname{ord}_{s_0} L(\pi, s)$. It is *admissible* if $\langle n, X \rangle \geq 0$ for every X in the class of (F2).

The true orders at any s_0 form an admissible assignment, because every $L(X, s)$ with X in that class is holomorphic (Hecke, Langlands–Tunnell), and the class is closed under the functoriality listed in (F2).

Proposition 3 ([T]). *Let $s_0 = \frac{1}{2} + i\gamma$ be a zero of $L(6 \otimes \chi^3, s)$. The assignment $n = \theta^*$ is admissible. It has $n_\rho = -1$, $n_{6 \otimes \chi^3} = 1$ and all other $n_\pi = 0$. It is compatible with all of the following:*

- *Artin's conjecture for every irreducible outside the twist class of ρ ;*
- *holomorphy of every $L(X, s)$ for X in the class of (F2);*
- *GRH for every entire L -function involved, and for $L(\rho)$ in the meromorphic sense (zeros and poles on the line).*

The splitting $e_{6 \otimes \chi^3} = \theta^* + e_\rho$, with both summands admissible, places a simple pole of $L(\rho)$ at s_0 and a zero of $L(\rho)$ at a point s_1 on the line, with s_1 as close to s_0 as desired. This leaves the zero counts of every entire L -function attached to G unchanged in any disc containing s_0 and s_1 .

Proof. Admissibility is (F2). The assignment involves only the points s_0 and s_1 , both on the critical line, so it is compatible with GRH. It changes no order of any $L(X, s)$ with X in the class, beyond what θ^* and e_ρ do separately, and both pair non-negatively with every such X . $L(6 \otimes \chi^3)$ has infinitely many zeros on the line, so the configuration can be repeated at infinitely many heights, consistently with Booker's theorem. $\square$

Remark 4. Proposition 3 is a statement about bookkeeping. It does not construct a counterexample function. What it shows is that GRH moves the question onto the critical line and stops there. A proof of Artin's conjecture for ρ from GRH would need further input about *which* points on the line are zeros of which functions. That is equality information, and it is exactly what numerical enclosures cannot certify (companion paper, Remark 13.5).

4 Artin's conjecture for ρ under Selberg's orthonormality conjecture

The Selberg class. $\mathcal{S}$ is the class of Dirichlet series $F(s) = \sum a_F(n)n^{-s}$ with the following properties:

- they converge absolutely for $\operatorname{Re} s > 1$;
- $(s-1)^m F(s)$ extends to an entire function of finite order for some m ;
- they satisfy a functional equation of the standard shape and the Ramanujan bound $a_F(n) \ll n^\varepsilon$;
- they have an Euler product $\log F(s) = \sum b_F(n)n^{-s}$ with b_F supported on prime powers and $b_F(n) \ll n^\vartheta$ for some $\vartheta < \frac{1}{2}$.

Note that $b_F(p) = a_F(p)$. A nonconstant $F \in \mathcal{S}$ is *primitive* if $F = GH$ with $G, H \in \mathcal{S}$ forces $G = 1$ or $H = 1$. Every $F \in \mathcal{S}$ is a finite product of primitive elements [2], because the degree is additive and there are no elements of degree strictly between 0 and 1. Hecke L -functions of finite-order Hecke characters lie in $\mathcal{S}$.

Conjecture B (Selberg). For primitive $F, G \in \mathcal{S}$,

$$\sum_{p \leq x} \frac{a_F(p) \overline{a_G(p)}}{p} = \delta_{F,G} \log \log x + O(1).$$

Lemma 5 (unique factorisation under B). *Assume Conjecture B. Let $F_1, \dots, F_r$ be distinct primitive elements and $e_j \in \mathbb{Z}$. If $\prod_j F_j^{e_j} = 1$ as meromorphic functions, then every $e_j = 0$.*

Proof. Taking logarithms and the coefficient of p^{-s} gives $\sum_j e_j a_{F_j}(p) = 0$ for every prime p . Multiply by $\overline{a_{F_k}(p)}/p$ and sum over $p \leq x$. Conjecture B gives $e_k \log \log x = O(1)$, so $e_k = 0$. $\square$

Lemma 6 (Chebotarev, unconditional). *For a character ξ of G ,*

$$\sum_{p \leq x, p \nmid 1951} \frac{|\xi(\operatorname{Frob}_p)|^2}{p} = \langle \xi, \xi \rangle \log \log x + O(1).$$

Proof. Write $|\xi|^2 = \sum_C |\xi(C)|^2 \mathbf{1}_C$ over conjugacy classes C . The Chebotarev density theorem with its classical error term gives

$$\sum_{p \leq x, \text{Frob}_p \in C} \frac{1}{p} = \frac{|C|}{|G|} \log \log x + c_C + o(1),$$

by partial summation. Sum over C and use $\sum_C |C| |\xi(C)|^2 / |G| = \langle \xi, \xi \rangle$. $\square$

Theorem 7 ([T|Conjecture B]). *Assume Selberg's Conjecture B. Then $L(\rho, s)$ is entire. More generally, $L(\pi, s)$ is entire for every nontrivial irreducible π of G , and each such $L(\pi, s)$ is a primitive element of $\mathcal{S}$.*

Proof. Step 1: a single primitive factor. Let π be a nontrivial irreducible character of G . By Brauer's induction theorem, $\pi = \sum_i n_i \text{Ind}_{H_i} \psi_i$ with $n_i \in \mathbb{Z}$ and ψ_i linear, so $L(\pi, s) = \prod_i L(K_i, \psi_i, s)^{n_i}$, a quotient of elements of $\mathcal{S}$. Factor every $L(K_i, \psi_i)$ into primitives and collect the result: $L(\pi, s) = \prod_j F_j^{e_j}$ with distinct primitive F_j and $e_j \in \mathbb{Z}$. Comparing coefficients of p^{-s} in the logarithms gives $\pi(\text{Frob}_p) = \sum_j e_j a_{F_j}(p)$ for $p \nmid 1951$. Lemma 6 with $\langle \pi, \pi \rangle = 1$ gives

$$\log \log x + O(1) = \sum_{p \leq x} \frac{|\pi(\text{Frob}_p)|^2}{p} = \sum_{j,k} e_j \overline{e_k} \sum_{p \leq x} \frac{a_{F_j}(p) \overline{a_{F_k}(p)}}{p} = \left(\sum_j e_j^2 \right) \log \log x + O(1),$$

where the last step is Conjecture B. Hence $\sum_j e_j^2 = 1$, so $L(\pi, s) = F_\pi^{\varepsilon_\pi}$ for one primitive F_π and some $\varepsilon_\pi \in \{\pm 1\}$.

Step 2: different representations give different factors. If $F_\pi = F_{\pi'}$, then $\varepsilon_\pi \pi(\text{Frob}_p) = \varepsilon_{\pi'} \pi'(\text{Frob}_p)$ for all $p \nmid 1951$. By Chebotarev every conjugacy class is a Frobenius class, so $\pi = \pm \pi'$ as class functions. Evaluating at the identity gives the sign $+$, so $\pi = \pi'$.

Step 3: the sign is $+1$. The Dedekind zeta function satisfies $\zeta_N = \prod_\pi L(\pi, s)^{\pi(1)}$, so $\zeta_N = \zeta \cdot \prod_{\pi \neq 1} F_\pi^{\pi(1)\varepsilon_\pi}$. Since $\zeta_N \in \mathcal{S}$, it is also a product of primitives with non-negative exponents. The Riemann zeta function is primitive in $\mathcal{S}$, and by Step 2 the F_π are distinct; $F_\pi \neq \zeta$ because $\pi \neq 1$, by the argument of Step 2. Lemma 5 shows the two factorisations agree, so $\pi(1)\varepsilon_\pi \geq 0$, that is $\varepsilon_\pi = +1$.

Step 4: entire. So $L(\pi, s) = F_\pi \in \mathcal{S}$, and $(s-1)^m L(\pi, s)$ is entire. Classically, by Brauer and the non-vanishing of Hecke L -functions at $s=1$, $L(\pi, s)$ is holomorphic and non-zero at $s=1$ for nontrivial π . Hence $L(\pi, s)$ is entire. Taking $\pi = \rho$ gives the theorem. $\square$

Remark 8 (how this gets past the barrier). Conjecture B is not a statement about orders of zeros, so Proposition 3 does not apply to it. It controls the prime coefficients of the unknown primitive factors. For ρ these coefficients are known exactly (Frobenius traces), and their mean square is 1 (Lemma 6). In the configuration of Proposition 3 the factorisation of $L(\rho)$ would have to contain a primitive factor to a negative power. Step 3 rules that out using the positivity of the exponents in ζ_N . So the barrier is broken by an input of exactly the kind it predicts: information attached to ρ itself, namely the mean square of its Frobenius traces against the primitive factors of the Hecke L -functions.

Remark 9 (how far Conjecture B is known). Selberg orthogonality is a theorem for cuspidal automorphic L -functions of GL_m in many cases [4]. The difficulty here is that the primitive factors of the Hecke L -functions in Step 1 are not known to be automorphic. Their automorphy is equivalent to Artin's conjecture for the corresponding irreducibles. What is actually needed is far weaker than Conjecture B for all of $\mathcal{S}$: see Section 5, where orthonormality among the primitive divisors of L_{12} and L_{10} (total degree 22) is shown to suffice.

5 A sharper conditional proof: two explicit Hecke L -functions suffice

Theorem 7 uses Conjecture B for all of $\mathcal{S}$. For ρ alone, much less is needed. Step 3 can also be replaced by a degree count, so ζ_N is not needed at all.

Hypothesis B(1951). Let $\mathcal{P}$ be the set of primitive $F \in \mathcal{S}$ that divide L_{12} or L_{10} in $\mathcal{S}$, that is, $L_{12}/F \in \mathcal{S}$ or $L_{10}/F \in \mathcal{S}$. For $F, G \in \mathcal{P}$,

$$\sum_{p \leq x} \frac{a_F(p) \overline{a_G(p)}}{p} = \delta_{F,G} \log \log x + O(1).$$

This is Conjecture B restricted to the primitive divisors of two explicit Hecke L -functions of total degree 22. It is a hypothesis about whatever those divisors are. It does *not* reduce to Chebotarev. If Artin's conjecture held for ρ and for $\text{Sym}^3 \rho_0 \otimes \chi^3$, and the three L -functions below were primitive, then $\mathcal{P} = \{L(\rho), L(\text{Sym}^3 \rho_0 \otimes \chi^3), L(6 \otimes \chi^3)\}$, and B(1951) would follow from Lemma 6 and the orthogonality of characters. Unconditionally, however, Chebotarev gives orthogonality of the *Frobenius traces* only, not membership in $\mathcal{S}$:

- $L(6 \otimes \chi^3)$ lies in $\mathcal{S}$, by (F3), but is not known to be primitive;
- $L(\rho) \in \mathcal{S}$ is the conclusion of Theorem 11;
- $L(\text{Sym}^3 \rho_0 \otimes \chi^3)$ is not known to be entire. The representation $4 \otimes \chi^3$ is not monomial: that would need a subgroup $H \leq G$ of index 4, so of order 150. Since $150 \nmid 120$, $H \supseteq 1 \times C_5$ and its image in $\text{SL}(2, 5)$ has order 30. That image contains the unique involution $-I$, so its image in A_5 has order 15, and A_5 has no subgroup of order 15. Moreover symmetric-cube functoriality applies to automorphic ρ_0 , which is not known.

So the expected three are not known to form $\mathcal{P}$.

Lemma 10 ([T]). *There is no $F \in \mathcal{S}$ with $F(s)L(\rho, s) = 1$.*

Proof. Write $\Phi(s) = Q^s \prod_j \Gamma(\lambda_j s + \mu_j) F(s)$ for the completed F , with degree $d_F = 2 \sum_j \lambda_j \geq 0$, and $\Lambda(\rho, s) = q^{s/2} \Gamma_{\mathbb{R}}(s)^2 L(\rho, s)$; ρ is even, so both archimedean parameters are 0. Both satisfy functional equations, $\Phi(s) = \omega \overline{\Phi(1 - \bar{s})}$ and $\Lambda(\rho, s) = W \overline{\Lambda(\bar{\rho}, 1 - s)}$. Also $\bar{F}(s)L(\bar{\rho}, s) = 1$, by conjugating coefficients. If $FL(\rho) = 1$, then the product $\gamma(s) := \Phi(s)\Lambda(\rho, s)$ is a product of exponentials and Gamma factors, of total degree $d = d_F + 2 \geq 2$, and it satisfies $\gamma(s) = \omega W \overline{\gamma(1 - \bar{s})}$. Stirling's formula gives $\log |\gamma(\sigma + it)| - \log |\gamma(1 - \sigma + it)| = d(\sigma - \frac{1}{2}) \log |t| + O_\sigma(1)$ as $|t| \rightarrow \infty$. The left side is constant, so $d = 0$, a contradiction. $\square$

Theorem 11 ([T|B(1951)]). *Assume Hypothesis B(1951). Then $L(\rho, s)$ is entire and primitive in $\mathcal{S}$, so Artin's conjecture holds for ρ .*

Proof. $L_{12}, L_{10} \in \mathcal{S}$ are Hecke L -functions. Each is a finite product of primitive elements [2], all of which lie in $\mathcal{P}$. By (F1), $L(\rho) = L_{12}/L_{10}$. Cancelling equal factors gives $L(\rho, s) = \prod_j F_j^{e_j}$ with distinct $F_j \in \mathcal{P}$ and $e_j \in \mathbb{Z}$. For $p \nmid 1951$ the coefficient of p^{-s} in $\log L(\rho, s)$ is $\rho(\text{Frob}_p)$, so $\rho(\text{Frob}_p) = \sum_j e_j a_{F_j}(p)$. Lemma 6 and B(1951) give, exactly as in Step 1 of Theorem 7, $\log \log x + O(1) = (\sum_j e_j^2) \log \log x + O(1)$. Hence $L(\rho) = F^{\pm 1}$ for one $F \in \mathcal{P}$. The exponent -1 is excluded by Lemma 10. So $L(\rho) = F \in \mathcal{S}$ and $(s-1)^m L(\rho, s)$ is entire. Finally, L_{12} and L_{10} are

Hecke L -functions of nontrivial characters, so they are holomorphic and non-zero at $s = 1$. Hence $L(\rho)$ is holomorphic at $s = 1$, and so $L(\rho)$ is entire. Since $L(\rho)$ then has no poles, Booker's theorem [1] also gives the corresponding automorphic form (strong Artin). $\square$

Remark 12 (the barrier again). In the configuration of Proposition 3, $L(\rho)$ has a pole at a zero of $L(6 \otimes \chi^3)$, so $L(\rho)$ is not in $\mathcal{S}$. Then its factorisation $\prod_j F_j^{e_j}$ must contain some $e_j < 0$. Either it is a single F^{-1} , which Lemma 10 excludes, or $\sum_j e_j^2 \geq 2$. B(1951) forbids the second case because ρ 's Frobenius traces have mean square 1. So the decisive input is coefficient information about the primitive divisors of L_{12} and L_{10} . Locations of zeros do not enter.

6 Summary

hypothesis	conclusion for ρ
none	poles lie in $1 - \frac{1}{8912+1145 \log t } > \operatorname{Re} s > \frac{1}{8912+1145 \log t }$ for $ t \geq 1$, at zeros of L_{10}
GRH	all poles on $\operatorname{Re} s = \frac{1}{2}$ at zeros of L_{10} ; either Artin or infinitely many such poles (Thm. 1)
GRH and Artin for every other irreducible, and all known functoriality	still not enough (Prop. 3)
Selberg's Conjecture B	Artin's conjecture for every irreducible of G , in particular ρ (Thm. 7)
B(1951): orthonormality for the primitive divisors of L_{12}, L_{10}	Artin's conjecture for ρ (Thm. 11)

Artin's conjecture for ρ remains open: GRH, Conjecture B and Hypothesis B(1951) are all unproved.

References

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